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National Cybersecurity Architecture

A Defense-in-Depth Reference Design

Carrier-Grade Traffic Intelligence (Allot) + Next-Generation Firewall Perimeter (Palo Alto Networks)

Field	Value
Prepared for	National Cybersecurity Authority
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Document Type	Solution Architecture & Advisory Whitepaper
Version	1.0
Issue Date	April 2026
Classification	Restricted — For Internal Steering Committee Review

Submitted to the National Cybersecurity Steering Committee



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1. Executive Summary

This document presents a consultant-grade reference architecture for safeguarding national digital infrastructure against the contemporary threat landscape. The design is engineered around a two-vendor, multi-layer defense-in-depth model that places carrier-grade traffic intelligence and security from Allot at the network backbone and citizen-facing layers, complemented by Palo Alto Networks Next-Generation Firewalls (NGFW) at the perimeter of government, defense, and critical-infrastructure enclaves.

The proposed architecture addresses six national security imperatives: (i) volumetric DDoS resilience for sovereign internet capacity, (ii) lawful regulatory enforcement on illegal and harmful content, (iii) protection of citizens — particularly children and small businesses — from phishing, malware, and IoT-borne threats, (iv) hardened perimeter security for ministries and critical sectors, (v) deep visibility into traffic and threat telemetry feeding a National Cybersecurity Operations Centre (N-CSOC), and (vi) policy-driven traffic management to preserve quality-of-service for essential digital public services.

Allot platforms — including the SG-Tera II family with up to 500 Gbps per chassis and 4 Tbps clustered throughput, supporting up to 4 million subscribers and 130 million concurrent IP flows — are positioned at internet gateways, peering exchanges, and ISP cores. Palo Alto NGFWs, scaling from the PA-400 series for branch enclaves to the PA-7000 series for data centres, are positioned at every critical perimeter where granular Layer-7 policy, sandboxing, and zero-trust segmentation are required.

The result is a layered, sovereign, and operationally cohesive cybersecurity posture that meets regulatory mandates, defends national capacity at line rate, and protects the digital experience of every connected citizen.

1.1 Key Outcomes

- **Sovereign DDoS shield** delivers sub-second mitigation at international gateways.
- **Nationwide content-compliance and parental-control fabric** without endpoint installation friction.
- **IoT botnet containment** with smart quarantine, breaking the kill chain before exfiltration.
- **Zero-trust perimeter** for every ministry, critical-information-infrastructure (CII) operator, and state data centre.
- **Unified telemetry pipeline** feeding the N-CSOC for real-time situational awareness.



2. Document Purpose & Scope

2.1 Purpose

This whitepaper serves as a strategic and technical advisory deliverable for the steering committee responsible for chartering, funding, and governing the national cybersecurity programme. It articulates the architecture, vendor positioning, capability mapping, capacity planning, cost drivers, and phased delivery roadmap necessary to convert policy intent into deployable capability.

2.2 In Scope

- National internet gateway and ISP core security (volumetric protection, content filtering, lawful intercept enablement).
- Subscriber-facing protection for residential, SMB, and IoT users.
- Perimeter security for government enclaves, ministries, and critical-information-infrastructure operators.
- Centralised telemetry, analytics, and policy management.
- Operational model and governance for a National Cybersecurity Operations Centre.

2.3 Out of Scope

- Endpoint detection and response (EDR), email gateway, and identity provider selection — addressed in companion documents.
- Procurement legal terms, country-specific data residency law interpretation, and political negotiations with vendors.
- Application-layer software security testing of government applications.

2.4 Audience

This document is written for the National Cybersecurity Coordinator, the CISO community of state-owned enterprises and ministries, the leadership of telecom regulators and licensed ISPs, and the technical evaluation committee. Readers are assumed to have a working familiarity with TCP/IP, NGFW concepts, and operator-grade networks.

3. National Threat Landscape

The threats relevant to this architecture span the full spectrum from low-effort opportunistic attacks against citizens to nation-state-grade campaigns targeting critical infrastructure. The architecture is designed to neutralise or contain each of the following classes.



Threat Class	National-Level Impact	Likelihood / Trend
Volumetric & Reflective DDoS	Saturate national or sectoral internet capacity to degrade public services.	High — recurrent during geopolitical events
Botnets & IoT Worms	Compromised consumer routers, cameras, and DVRs weaponised for outbound attacks.	High — millions of vulnerable consumer endpoints
Phishing & Credential Theft	Mass campaigns harvesting banking, government-ID, and OTT credentials.	Very High — primary entry vector
Ransomware (Targeted)	Lateral movement into ministries, hospitals, and utilities.	High — rising trajectory
Advanced Persistent Threat (APT)	Long-dwell espionage against defence, foreign affairs, and energy.	Critical — strategic impact
Illegal & Harmful Content	Distribution of CSAM, terror propaganda, and other prohibited material.	Persistent — regulatory mandate
Data Exfiltration	Bulk movement of citizen data via DNS tunnels, encrypted channels, or insiders.	High — sovereignty concern
Supply-Chain & Zero-Day	Exploitation of unknown vulnerabilities in widely deployed software.	High — low predictability

The architecture in Section 5 maps each of these classes to specific platform capabilities, ensuring no class is left without an explicit control.

4. Strategic Objectives & Design Principles

4.1 Strategic Objectives

1. Preserve national internet availability under volumetric attack.
2. Protect every connected citizen by default, without requiring software installation.
3. Enforce regulatory and judicial directives on prohibited content uniformly across licensed operators.
4. Harden the perimeter of government, defence, and critical-infrastructure operators to internationally recognised NGFW standards.
5. Deliver unified, real-time situational awareness to the N-CSOC.
6. Establish a sustainable operating model with measurable KPIs and clear escalation.



4.2 Design Principles

- **Defense in Depth:** No single control point is trusted; protections are layered from the international gateway down to the device.
- **Right Tool, Right Layer:** Carrier-grade DPI and DDoS at the backbone; granular Layer-7 NGFW at perimeters. Avoid forcing either platform outside its design envelope.
- **Zero User Friction:** Citizen-facing protection is delivered network-side wherever possible, removing the burden of installation and updates.
- **Zero Trust at the Edge:** Within enclaves, every flow is authenticated, inspected, and policy-evaluated regardless of origin.
- **Sovereign Data Handling:** Telemetry, threat intelligence, and citizen metadata are stored, processed, and governed in-country.
- **Operationalised, Not Just Installed:** Every control has a corresponding playbook, KPI, and named owner in the N-CSOC.
- **Modularity & Phased Investment:** Capability rolls out in waves so value is realised early and budget is consumed against measurable outcomes.

5. Reference Architecture

The architecture is organised into four concentric protection rings, each anchored on the platform best suited to its scale, latency, and policy requirements.

5.1 Logical Architecture Overview

Ring	Where Deployed	Recommended Platform	Primary Controls
Ring 1 — National Backbone	International gateways, IXPs, ISP cores	Allot SG-Tera II / SG-9000 (Smart NetProtect, Traffic Intelligence)	Volumetric DDoS, illegal content filtering, traffic visibility, regulatory compliance
Ring 2 — Subscriber Edge	Mobile / fixed broadband BNGs and aggregation	Allot NetworkSecure, IoTSecure, HomeSecure	Anti-phishing, anti-malware, parental control, IoT botnet containment
Ring 3 — Sector Perimeter	Ministries, defence, finance, healthcare, energy	Palo Alto PA-5400 / PA-7000 series NGFW	App-ID, User-ID, Threat Prevention, WildFire, URL Filtering, DNS Security
Ring 4 — Branch & Enclave	Government district offices, hospitals, schools, substations	Palo Alto PA-400 / PA-1400 series NGFW	Zero-trust segmentation, IoT/OT identification, encrypted-traffic inspection



5.2 Ring 1 — National Backbone Protection (Allot)

At national peering points and the cores of every licensed ISP, Allot Smart NetProtect provides inline, zero-day DDoS mitigation. Host Behaviour Anomaly Detection (HBAD) identifies infected hosts and blocks volumetric and reflective attacks within seconds, well below the SLA window typically required to keep public services online during a campaign.

Allot Traffic Intelligence runs alongside, providing Deep Packet Inspection (DPI), application-aware analytics, and quality-of-service enforcement for essential digital public infrastructure (DPI) services such as identity, payments, and emergency communications. The same platform supports illegal-content blacklist enforcement under judicial or regulatory direction, ensuring uniform compliance across all licensed operators.

- **Recommended chassis:** SG-Tera II (up to 500 Gbps per chassis, clusterable to 4 Tbps) at Tier-1 gateways; SG-9000 series at Tier-2 ISP cores.
- **Scale envelope:** Up to 4 million subscribers and 130 million concurrent IP flows per system — sufficient for a mid-to-large nation's Tier-1 operator.
- **Deployment mode:** Inline, transparent, with bypass for fail-safe operation. NFV / COTS deployment available where virtualisation policy mandates.

5.3 Ring 2 — Subscriber Protection (Allot Secure portfolio)

Allot NetworkSecure delivers personal security, parental control, and ad-blocking as a service-provider feature, requiring no installation on the subscriber's device. This is decisive at national scale: it eliminates the single largest barrier to citizen cybersecurity adoption — the requirement that every individual download, configure, and maintain client software.

For households, Allot HomeSecure adds a thin Customer Premises Equipment (CPE) agent that hardens router passwords, closes open ports, and monitors traffic for indicators of compromise. For the explosive growth of consumer and municipal IoT — cameras, smart meters, traffic sensors, set-top boxes — Allot IoTSecure profiles device behaviour, detects anomalies, and applies smart quarantine to infected devices, breaking botnet command-and-control before outbound damage occurs.

- **Citizen value:** Phishing, malware, and harmful-content protection delivered from the network — opt-in or opt-out per regulatory model.
- **Operator value:** New ARPU stream, reduced abuse-desk load, and reduced upstream bandwidth consumed by botnet traffic.
- **National value:** Massive reduction in the population of compromised devices that could be weaponised against domestic or foreign targets.



5.4 Ring 3 — Sector & Government Perimeter (Palo Alto NGFW)

Every ministry, defence command, central bank, stock exchange, healthcare network, energy operator, and state data centre is treated as a distinct security enclave. At the perimeter of each, Palo Alto Networks NGFWs enforce identity-aware, application-aware, and content-aware policy.

- **App-ID and User-ID:** Policy is written in business terms — by application and identity — rather than IP and port, dramatically reducing rule-base sprawl and audit complexity.
- **Threat Prevention and WildFire:** Inline IPS plus cloud sandboxing for zero-day malware. WildFire telemetry feeds back into a national threat-intelligence pool curated by the N-CSOC.
- **URL Filtering and DNS Security:** Defence against phishing, command-and-control beaconing, and DNS-tunnel exfiltration.
- **Data Filtering:** Detection and blocking of sensitive-data exfiltration patterns — citizen identifiers, health records, financial data — supporting GDPR-style and sectoral compliance.
- **Recommended models:** PA-7000 series for national data centres; PA-5400 series for ministry headquarters; PA-3400 / PA-1400 for medium enclaves.

5.5 Ring 4 — Branch, Field & OT Protection (Palo Alto NGFW)

District offices, hospitals, schools, electrical substations, and water-treatment facilities receive PA-400 series NGFWs. These provide zero-trust segmentation between IT and OT zones, identification of unmanaged IoT/OT devices, and inspection of encrypted traffic — closing the most common lateral-movement paths exploited in modern targeted intrusions.

5.6 Cross-Cutting: Telemetry, Analytics, and the N-CSOC

Both vendors expose rich telemetry. Allot's Traffic Intelligence and ACTI feed flow records, application analytics, and threat events. Palo Alto NGFWs forward logs, threat events, and WildFire verdicts. All telemetry is normalised into the National Cybersecurity Operations Centre's SIEM and SOAR platform, where correlation rules, threat-hunting playbooks, and automated response are executed by a 24×7 staff.

6. Capability Mapping: Threat-to-Control

The matrix below makes explicit which platform addresses which threat class, removing ambiguity in vendor positioning and ensuring no gap in coverage.

Threat / Requirement	Allot Coverage	Palo Alto Coverage
Volumetric / Reflective DDoS	Allot Smart NetProtect (HBAD, sub-second inline mitigation)	Palo Alto Zone Protection & DoS Profiles (enclave-level)
Botnets / IoT Worms	Allot IoTSecure (smart quarantine, behaviour profiling)	Palo Alto Botnet Detection (compromised-host alerts)



Threat / Requirement	Allot Coverage	Palo Alto Coverage
Phishing & Malicious URLs	Allot NetworkSecure (network-side URL reputation)	Palo Alto URL Filtering (PAN-DB, custom policies)
Malware & Ransomware	Allot NetworkSecure / HomeSecure (signature + reputation)	Palo Alto Threat Prevention + WildFire (zero-day sandbox)
Illegal / Harmful Content	Allot DPI-based Illegal Content Filtering (operator-wide)	Palo Alto Custom Blocklists & URL Filtering (enclave-wide)
Data Exfiltration	Allot Traffic Intelligence (anomalous-flow analytics)	Palo Alto Data Filtering + DNS Security
APT Lateral Movement	Out of scope for backbone tier	Palo Alto NGFW segmentation, App-ID, User-ID, Threat Prevention
Parental Control / Child Safety	Allot NetworkSecure (per-subscriber profiles)	Palo Alto URL Filtering (institutional networks: schools, libraries)
IoT / OT Visibility	Allot IoTSecure (consumer / municipal IoT)	Palo Alto Device-ID (industrial OT and enterprise IoT)
Quality-of-Service for DPI Services	Allot Traffic Intelligence (policy-based bandwidth)	Out of scope for NGFW tier

6.1 Why Two Vendors and Not One

A single-vendor approach is operationally simpler but architecturally weaker. Carrier-grade DPI and DDoS at terabit scale, and granular Layer-7 NGFW with sandboxing at perimeters, are different engineering disciplines optimised by different vendors. Forcing either platform outside its design envelope creates either a capacity ceiling or a policy-granularity ceiling, both of which are unacceptable at national scale. The two platforms also create a deliberate, defensible vendor diversity that reduces systemic risk from any single supply-chain compromise.

7. Capacity Planning & Sizing Guidance

7.1 Allot Sizing Inputs

Allot platforms are sized against four primary inputs: aggregate throughput in Gbps, concurrent IP flows, subscriber count, and feature mix (DPI alone, DPI + DDoS, DPI + DDoS + NetworkSecure).

Allot Platform	Indicative Throughput	Typical Position	Feature Envelope
SSG 200 / 400	Up to ~10 Gbps	Tier-3 ISP, regional aggregation	DPI + basic DDoS



Allot Platform	Indicative Throughput	Typical Position	Feature Envelope
SSG 600 / 800	Up to ~40 Gbps	Tier-2 ISP core	DPI + DDoS + NetworkSecure
SG-Sigma E6 / E14	40 – 200 Gbps	Large Tier-2 / small Tier-1	Full security suite
SG-9000 Series	100 – 300 Gbps	Tier-1 ISP regional	Full security suite
SG-Tera II	Up to 500 Gbps single, 4 Tbps clustered	International gateway, IXP	Full security suite at carrier scale

As a planning rule of thumb, peak observed traffic at a national gateway is sized at 1.6× the busy-hour average, with an additional 30% headroom for DDoS surge absorption. A nation with 6 Tbps of busy-hour international traffic should therefore plan for approximately 12.5 Tbps of inspection capacity, achievable with three clustered SG-Tera II deployments distributed across geographically diverse landing stations.

7.2 Palo Alto Sizing Inputs

NGFW sizing is driven by Threat-Prevention-enabled throughput (not raw firewall throughput), session count, new-connections-per-second, and the SSL/TLS decryption ratio. Modern enclaves see 80%+ encrypted traffic; sizing must assume decryption is on for almost every flow.

Palo Alto Series	Typical Deployment	Indicative Threat-Prevention Throughput
PA-400 Series	Branch office, small enclave	1 – 3 Gbps Threat Prevention
PA-1400 Series	Medium ministry office, district HQ	5 – 10 Gbps Threat Prevention
PA-3400 Series	Large enclave, mid-tier data centre	10 – 25 Gbps Threat Prevention
PA-5400 Series	Ministry HQ, sectoral data centre	25 – 100 Gbps Threat Prevention
PA-7000 Series	National data centre, internet edge of large operator	100 – 700 Gbps Threat Prevention

All NGFWs in the design are deployed in active-active or active-passive HA pairs. Panorama is mandatory at this scale to centralise policy authoring, log aggregation, and version control across the fleet.

8. Cost & Total-Cost-of-Ownership Drivers

Procurement at national scale requires the steering committee to negotiate against a clear understanding of what actually drives spend. Headline list prices are misleading; the variables below dominate true TCO over a five-to-seven-year horizon.



8.1 Allot Cost Drivers

- **Throughput tier:** Pricing scales with Mbps / Gbps inspection capacity; carrier-class platforms (SG-Tera II) carry the highest unit cost.
- **Deployment model:** Physical appliance (NetEnforcer, SSG) versus virtualised NFV on COTS hardware. NFV typically reduces hardware capex but shifts cost to compute, storage, and orchestration licensing.
- **Feature licences:** Traffic management, DDoS protection, NetworkSecure, IoTSecure, and analytics modules are licensed independently. License only what each site needs.
- **Subscriber-tier licensing:** NetworkSecure-class features price by subscriber count; volume discounting applies above defined thresholds.
- **Support tier:** 24x7 with on-site spares is mandatory at backbone tier; lower tiers may suffice at lower rings.

8.2 Palo Alto Cost Drivers

- **Hardware model:** Costs scale with the platform series, from PA-400 at the branch to PA-7000 at the national data centre.
- **Security subscriptions:** Threat Prevention, WildFire, URL Filtering, and DNS Security are licensed individually. The Advanced bundles (Advanced Threat Prevention, Advanced WildFire, Advanced URL Filtering, Advanced DNS Security) increase efficacy but raise per-unit subscription cost.
- **Panorama and logging:** Centralised management and log retention drive substantial hidden cost; plan storage against retention-period regulatory requirements.
- **SSL decryption capacity:** Decryption can consume 60–70% of a chassis's rated Threat-Prevention throughput. Always size against decrypted-throughput, not raw firewall throughput.
- **Premium support:** 4-hour response with on-site sparing is required for critical-infrastructure deployments.

8.3 TCO Composition (Indicative)

Cost Category	Share of 5-Year TCO	Notes
Hardware capex	30 – 40%	Front-loaded; depreciate over 5 years
Subscription / feature licences	25 – 35%	Annual, recurring
Support & maintenance	10 – 15%	Annual, often bundled with subscriptions
Implementation services	5 – 10%	One-time; do not under-invest here
Operations (N-CSOC staff, training)	10 – 20%	Recurring; the largest sustained line item

Indicative numbers are typical for nation-scale deployments and should be validated against vendor RFP responses. The single most common procurement error is under-investing in implementation services and



operational staffing — both of which determine whether the platform actually delivers the promised security outcome.

9. Governance, Operations & KPIs

9.1 Governance Model

A three-tier governance structure is recommended. A National Cybersecurity Steering Committee, chaired by the National Cybersecurity Coordinator, sets policy and approves budget. A Programme Management Office (PMO) executes delivery against the agreed scope. The N-CSOC operates the platforms 24×7, reporting weekly to the PMO and monthly to the Steering Committee.

9.2 Operating Model — N-CSOC

- **Tier 1 (Triage):** 24×7 monitoring of unified telemetry; first-pass classification and containment.
- **Tier 2 (Investigation):** Deep analysis, malware reverse-engineering, threat hunting against retained logs.
- **Tier 3 (Response & Engineering):** Incident command, forensic recovery, platform tuning, content authoring for SIEM and SOAR.
- **Threat Intelligence Cell:** Curates national IOC feed, federates with allied CERTs, drives proactive blocking across both vendor fleets.

9.3 Key Performance Indicators

KPI	Target	Source
DDoS mitigation time	< 5 seconds from detection to suppression	Allot Smart NetProtect telemetry
NGFW decryption coverage	> 90% of HTTPS in critical enclaves	Palo Alto Panorama report
Subscriber phishing-block rate	Trending up month-on-month	Allot NetworkSecure dashboards
Mean Time to Detect (MTTD)	< 30 minutes for high-severity events	SIEM
Mean Time to Respond (MTTR)	< 4 hours for high-severity events	SOAR / ticketing
Patch / signature freshness	< 24 hours from vendor release	Configuration audit
IoT smart-quarantine rate	100% of confirmed compromised devices contained	Allot IoTSecure
Unauthorised content block uptime	≥ 99.95% on regulated lists	Allot DPI compliance report



10. Risks & Mitigation

Every national-scale cybersecurity programme is constrained by a small set of recurring risks that, if left unmanaged, will erode the programme's outcomes long before its budget is exhausted. The matrix below captures the most material risks together with the mitigations that have been embedded into the architecture and the operating model.

Risk	Likelihood	Impact	Mitigation
Skills shortage in N-CSOC	High	High	Multi-year training contract with both vendors; accredited certifications; partnership with national universities for talent pipeline.
Vendor lock-in	Medium	Medium	Two-vendor architecture by design; data portability requirements written into contracts; logs forwarded to vendor-neutral SIEM.
Performance degradation under decryption load	Medium	High	Size NGFWs against decrypted throughput; selective decryption policy for low-risk categories; hardware refresh cycle every 4 years.
Regulatory drift / scope creep on filtering mandates	High	Medium	Formal change-control board for blacklist updates; auditability and judicial review of every entry.
Supply-chain compromise	Low	Critical	Hardware verification on receipt; firmware integrity monitoring; geographic supplier diversification.
Budget reduction in out-years	Medium	High	Phased delivery so Phase 1 value is irreversible; subscription licences renegotiated annually with measured outcome data.
Privacy / civil-liberties concerns over DPI	Medium	High	Independent oversight body; metadata-only processing where lawful; transparency reports.
Encrypted-traffic blind spots	High	Medium	Mandate decryption in critical enclaves; deploy NDR (out of current scope) for backbone observation.

11. Recommendations

Based on the analysis above, the following twelve recommendations are presented to the Steering Committee for endorsement. They are grouped by theme and tagged against their relative priority for the programme.



#	Theme	Recommendation	Priority
1	Architecture	Adopt the four-ring defense-in-depth architecture as the national reference design.	Foundational
2	Vendor Positioning	Position Allot at backbone and subscriber rings; position Palo Alto at sector and branch perimeters. Do not blur this division.	Foundational
3	Procurement	Issue a single, programme-level RFP per vendor track to maximise commercial leverage and ensure architectural coherence.	High
4	Backbone Defence	Deploy Allot SG-Tera II at every international gateway as a first priority; nothing else creates more national-security value per dollar in year one.	Critical
5	CII Perimeter	Mandate Palo Alto NGFW with Threat Prevention, WildFire, URL Filtering, and DNS Security as the baseline subscription bundle for all CII operators.	High
6	Citizen Protection	Roll out Allot NetworkSecure across licensed ISPs as a default-enabled, opt-out citizen protection service.	High
7	IoT Threat Surface	Treat IoT as a first-class threat surface — fund Allot IoTSecure ahead of the next election or major sporting event whose dates concentrate botnet risk.	High
8	Operations	Stand up the N-CSOC in parallel with platform deployment, not after; platforms without operators are theatre.	Critical
9	Policy Governance	Centralise policy in Panorama and Allot management; resist the temptation to allow each ministry to author its own NGFW policy.	Foundational
10	People & Skills	Invest 10–20% of the programme TCO in operations, training, and tabletop exercises; this is the highest-ROI line item.	Continuous
11	Civil Liberties	Establish independent civil-liberties oversight of all DPI and content-filtering activity from day one to retain public trust.	High
12	Transparency	Publish an annual transparency report covering DDoS events absorbed, malware blocked, and citizens protected — make the programme's value visible to those who fund it.	Recurring

12. Conclusion

Cybersecurity at national scale is no longer an IT discipline; it is an instrument of sovereignty, public safety, and economic continuity. The architecture proposed in this document combines two best-of-breed platforms, each deployed at the layer where it is engineered to excel, to deliver protection from the international gateway to the citizen's IoT camera.



Allot delivers the carrier-grade resilience, citizen-friendly protection, and regulatory enforcement that only platforms built for terabit traffic and tens of millions of subscribers can deliver. Palo Alto Networks delivers the granular, identity-aware, application-aware perimeter and segmentation control that the most sensitive enclaves of government and critical infrastructure require.

The architecture is designed to deliver visible value early, sustain political and budgetary support over time, and culminate in a mature, automated, and continuously improving national posture. With the governance, operations, and oversight structures recommended above, this programme is achievable, defensible, and worthy of national investment.

Rajendra Management Pvt. Ltd. and the Senior Solution Architect remain available to the Steering Committee for follow-up briefings, vendor RFP support, and design clarification.

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Appendix A — Glossary

Term	Definition
APT	Advanced Persistent Threat — long-dwell, well-resourced, often state-aligned intrusion campaign.
BNG	Broadband Network Gateway — aggregates subscriber sessions in a fixed-line operator network.
CII	Critical Information Infrastructure — networks whose disruption has national-level consequences.
CPE	Customer Premises Equipment — the device on the subscriber's premises (router, ONT).
DDoS	Distributed Denial of Service — attack designed to exhaust capacity of a target.
DPI	Deep Packet Inspection (network) / Digital Public Infrastructure (policy) — context-dependent.
HBAD	Host Behaviour Anomaly Detection — Allot's behavioural DDoS-source identification.
IXP	Internet Exchange Point — physical infrastructure where networks peer.
NFV	Network Function Virtualisation — running network functions as software on COTS hardware.
NGFW	Next-Generation Firewall — Layer-7 aware, identity-aware firewall.
N-CSOC	National Cybersecurity Operations Centre.
OT	Operational Technology — industrial control systems, SCADA.
QoS	Quality of Service — bandwidth and latency assurance for prioritised flows.
SOAR	Security Orchestration, Automation, and Response.
TCO	Total Cost of Ownership.



Appendix B — Document Control

Document Metadata

Field	Detail
Document Title	National Cybersecurity Architecture — A Defense-in-Depth Reference Design
Document Owner	Rajendra Management Pvt. Ltd. — Network Security Practice
Author	Senior Solution Architect, RMPL
Reviewer	Practice Lead, Cybersecurity Advisory
Approver	Director, Strategic Advisory
Version	1.0
Issue Date	April 2026
Next Review	October 2026
Distribution	National Cybersecurity Steering Committee — restricted
Classification	Restricted — Confidential

Version History

Version	Date	Author	Summary of Change
0.1	March 2026	RMPL Solution Architect	Initial outline and section structure.
0.5	April 2026	RMPL Solution Architect	Draft body content, threat-to-control matrix, sizing tables.
1.0	April 2026	RMPL Solution Architect	Roadmap, governance, branding, glossary; ready for Steering Committee.

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